Vector Algebra & Calculus

- $\nabla f = \hat{\boldsymbol{x}}\frac{\partial f}{\partial x} + \hat{\boldsymbol{y}}\frac{\partial f}{\partial y} + \hat{\boldsymbol{z}}\frac{\partial f}{\partial z}$
- $\nabla\cdot \boldsymbol{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$
- $\nabla\times \boldsymbol{F} = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$
- $\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a}\cdot\boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a}\cdot\boldsymbol{b})$
- $\nabla\times\nabla f = 0$
- $\nabla\cdot\nabla\times\boldsymbol{F} = 0$
- $\nabla^2 f = \nabla\cdot\nabla f$
- $\nabla\times\nabla\times\boldsymbol{F} = \nabla(\nabla\cdot\boldsymbol{F}) - \nabla^2\boldsymbol{F}$
- $\nabla^2\boldsymbol{F} = \hat{\boldsymbol{x}}\nabla^2 F_x + \hat{\boldsymbol{y}}\nabla^2 F_y + \hat{\boldsymbol{z}}\nabla^2 F_z$
- $\nabla\cdot(\boldsymbol{A}f) = f\nabla\cdot\boldsymbol{A} + \boldsymbol{A}\cdot\nabla f$
- $\nabla\times(\boldsymbol{A}f) = f\nabla\times\boldsymbol{A} + (\nabla f)\times\boldsymbol{A}$
- $\nabla\cdot(\boldsymbol{A}\times\boldsymbol{B}) = \boldsymbol{B}\cdot\nabla\times\boldsymbol{A} - \boldsymbol{A}\cdot\nabla\times\boldsymbol{B}$

Gauss Theorem: $\iiint_V \nabla\cdot\boldsymbol{A}\mathrm{d}V = \oiint_S \boldsymbol{A}\cdot\mathrm{d}\boldsymbol{S}$

Stokes Theorem: $\iint_S \nabla\times\boldsymbol{A}\cdot\mathrm{d}S = \oint_C \boldsymbol{A}\cdot\mathrm{d}\boldsymbol{l}$

Electrostatics

Coulomb's Law (in free space): $\boldsymbol{E}(\boldsymbol{r}) = \frac{Q}{4\pi\varepsilon_0 r^2}\hat{\boldsymbol{r}}$

Gauss's Law (in free space): $\oiint_S \boldsymbol{E}\cdot\mathrm{d}\boldsymbol{S} = \frac{1}{\varepsilon_0}\iiint \rho\mathrm{d}V$

Conservative / Irrotational: $\oint_C \boldsymbol{E}\cdot\mathrm{d}\boldsymbol{l} = 0$

Electrostatic potential V: $\boldsymbol{E} = -\nabla V$

Poisson Equation (in free space): $\nabla^2 V = -\frac{\rho}{\varepsilon_0}$

In free space: $V(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\iiint \frac{\rho(\boldsymbol{r'})}{|\boldsymbol{r}-\boldsymbol{r'}|}\mathrm{d}\boldsymbol{r'}$

Magnetostatics

Biot-Savart Law (in free space): $\mathrm{d}\boldsymbol{B} = \frac{\mu_0}{4\pi}\frac{I\boldsymbol{l}\times\hat{R}}{R^2}$
Ampere's Law (in free space): $\oint_C \boldsymbol{B}\cdot\mathrm{d}\boldsymbol{l} = \mu_0 I$
Gauss's Law: $\oiint_S \boldsymbol{B}\cdot\mathrm{d}\boldsymbol{S} = 0$
Magnetic vector potential $\boldsymbol{A}$: $\boldsymbol{B} = \nabla\times\boldsymbol{A}$
Poisson Equation (in free space): $\nabla^2\boldsymbol{A} = -\mu_0\boldsymbol{J}$
In free space: $\boldsymbol{A}(\boldsymbol{r}) = \frac{\mu_0}{4\pi}\iiint \frac{\boldsymbol{J}(\boldsymbol{r'})}{|\boldsymbol{r}-\boldsymbol{r'}|}\mathrm{d}\boldsymbol{r'}$

Maxwell's Equations

Maxwell's equations in integral form in free space:

$$\oint_C \boldsymbol{E}\cdot\mathrm{d}\boldsymbol{l} = -\frac{\mathrm{d}}{\mathrm{d}t}\iint_S \boldsymbol{B}\cdot\mathrm{d}\boldsymbol{S}, \text{ Faraday's Law (stationary loop)}$$

$$\oint_C \boldsymbol{B}\cdot\mathrm{d}\boldsymbol{l} = \mu_0\varepsilon_0\frac{\mathrm{d}}{\mathrm{d}t}\iint_S \boldsymbol{E}\cdot\mathrm{d}\boldsymbol{S} + \mu_0 I, \text{ Maxwell-Ampere's Law}$$

$$\oiint_S \boldsymbol{E}\cdot\mathrm{d}\boldsymbol{S} = \frac{Q}{\varepsilon_0}, \text{ Gauss's Law}$$

$$\oiint_S \boldsymbol{B}\cdot\mathrm{d}\boldsymbol{S} = 0, \text{ Gauss's Law - magnetic}$$

Faraday's law (equation) on a non-stationary loop:
$\mathcal{E} = \oint_C(\boldsymbol{E}+\boldsymbol{v}\times\boldsymbol{B})\cdot\mathrm{d}\boldsymbol{l} = -\frac{\mathrm{d}}{\mathrm{d}t}\iint_S \boldsymbol{B}\cdot\mathrm{d}\boldsymbol{S} = -\frac{\mathrm{d}\Psi}{dt}$
Maxwell's equations in differential form in material:

$$\nabla\cdot\boldsymbol{D} = \rho_v, \qquad \nabla\cdot\boldsymbol{B} = 0,$$
$$\nabla\times\boldsymbol{E} = -\frac{\partial\boldsymbol{B}}{\partial t}, \quad \nabla\times\boldsymbol{H} = \boldsymbol{J} + \frac{\partial\boldsymbol{D}}{\partial t}.$$

Current Continuity Equation

$$\nabla\cdot\boldsymbol{J} + \frac{\partial\rho}{\partial t} = 0$$

Constitutive relationship for linear material

$$\boldsymbol{D} = \varepsilon\boldsymbol{E} = \varepsilon_0\boldsymbol{E} + \boldsymbol{P}, \qquad \boldsymbol{P} = \varepsilon_0\chi_e\boldsymbol{E}$$
$$\boldsymbol{B} = \mu\boldsymbol{H} = \mu_0(\boldsymbol{H}+\boldsymbol{M}), \quad \boldsymbol{M} = \chi_m\boldsymbol{H}$$
$$\boldsymbol{J} = \sigma\boldsymbol{E}$$

perfect electric conductor (PEC): $\sigma = \infty$
perfect dielectric: $\sigma = 0$
diamagnetic materials: $\chi_m < 0$
paramagnetic materials: $\chi_m > 0$
ferromagnetic materials: $\chi_m \gg 1$

Polarization and Magnetization

dipole moment: $\boldsymbol{p} = q\boldsymbol{l}$
magnetic dipole moment: $\boldsymbol{m} = I\boldsymbol{s}$
bound charge density: $\rho_b = -\nabla\cdot\boldsymbol{P}$
polarization current: $\boldsymbol{J}_p = \frac{\partial\boldsymbol{P}}{\partial t}$
magnetization current: $\boldsymbol{J}_m = \nabla\times\boldsymbol{M}$

Lorentz-Drude models for material properties

Conductivity: $\sigma = \frac{Nq^2}{m(\nu+j\omega)}$
DC Susceptibility: $\chi_e = \frac{N_d e^2}{m\varepsilon_0\omega_0^2}$

Boundary Conditions

$$\hat{\boldsymbol{n}}\times(\boldsymbol{E}_1-\boldsymbol{E}_2) = 0, \qquad \hat{\boldsymbol{n}}\times(\boldsymbol{H}_1-\boldsymbol{H}_2) = \boldsymbol{J}_s,$$
$$\hat{\boldsymbol{n}}\cdot(\boldsymbol{D}_1-\boldsymbol{D}_2) = \rho_s, \qquad \hat{\boldsymbol{n}}\cdot(\boldsymbol{B}_1-\boldsymbol{B}_2) = 0.$$

$\hat{\boldsymbol{n}}$: interface normal pointing from region 2 to region 1

Capacitance, Conductance and Inductance

	Capacitance	**Conductance**	**Inductance**
Definition	$C = \frac{Q}{V}$	$G = \frac{I}{V}$	$L = \frac{\Phi}{I}$
Energy / Power	$U_E = \frac{1}{2}CV^2$	$P_d = VI$	$U_H = \frac{1}{2}LI^2$
Parallel Plate	$C = \varepsilon\frac{A}{d}$	$G = \sigma\frac{A}{d}$	$\mathcal{L} = \mu\frac{d}{A}$
Coaxial Cable	$\mathcal{C} = \varepsilon\frac{2\pi}{\ln(b/a)}$	$\mathcal{G} = \sigma\frac{2\pi}{\ln(b/a)}$	$\mathcal{L} = \mu\frac{\ln(b/a)}{2\pi}$
General Relation	$C = \varepsilon\cdot$G.F.	$G = \sigma\cdot$G.F.	$L = \mu/$G.F.

Energy, Power, Poynting's Theorem

Electric energy density: $u_E = \frac{1}{2}\boldsymbol{D}\cdot\boldsymbol{E} = \frac{1}{2}\varepsilon|\boldsymbol{E}|^2$
Magnetic energy density: $u_M = \frac{1}{2}\boldsymbol{B}\cdot\boldsymbol{H} = \frac{1}{2}\mu|\boldsymbol{H}|^2$
Electric energy: $U_E = \frac{1}{2}\iiint_V \rho_e V\mathrm{d}\boldsymbol{r} = \iiint_V u_E\mathrm{d}\boldsymbol{r}$
Magnetic energy: $U_M = \frac{1}{2}\iiint_V \boldsymbol{J}\cdot\boldsymbol{A}\mathrm{d}\boldsymbol{r} = \iiint_V u_M\mathrm{d}\boldsymbol{r}$

Poynting's Theorem in time domain:
$P_S = P_e + P_d + \frac{\partial}{\partial t}(W_m + W_e)$, where
$P_S = -\iiint_V \boldsymbol{E}\cdot\boldsymbol{J}_i d\boldsymbol{r}$: power supplied by the sources,
$P_e = \oiint_S(\boldsymbol{E}\times\boldsymbol{H})\cdot d\boldsymbol{S}$: power leaving through S,
$P_d = \iiint_V \sigma\boldsymbol{E}\cdot\boldsymbol{E}d\boldsymbol{r}$: power dissipated,
$W_m = \iiint_V \frac{1}{2}\mu\boldsymbol{H}\cdot\boldsymbol{H}d\boldsymbol{r}$: energy associated with $\boldsymbol{H}$,
$W_e = \iiint_V \frac{1}{2}\varepsilon\boldsymbol{E}\cdot\boldsymbol{E}d\boldsymbol{r}$: energy associated with $\boldsymbol{E}$

Complex Poynting's Theorem:
$P_S = P_e + \overline{P}_d + j2\omega(\overline{W}_m - \overline{W}_e)$, where
$P_S = -\frac{1}{2}\iiint_V \boldsymbol{E}\cdot\boldsymbol{J}_i^* d\boldsymbol{r}$: supplied complex power,
$P_e = \frac{1}{2}\oiint_S(\boldsymbol{E}\times\boldsymbol{H}^*)\cdot d\boldsymbol{S}$: exiting complex power

Wave Equation & Plane Waves

Wave equation: $\nabla^2\boldsymbol{E} - \frac{1}{v^2}\frac{\partial^2\boldsymbol{E}}{\partial t^2} = 0$, where $v = \frac{1}{\sqrt{\mu\varepsilon}}$
d'Alembert wave solution: $E_x(z,t) = Af(t-\frac{z}{v}) + Bf(t+\frac{z}{v})$

Helmholtz equation: $\nabla^2\boldsymbol{E} + k^2\boldsymbol{E} = 0$, where
$k = \omega\sqrt{\mu\varepsilon}$, $\eta = \sqrt{\frac{\mu}{\varepsilon}}$
Helmholtz equation in lossy material:
$\nabla^2\boldsymbol{E} - \gamma^2\boldsymbol{E} = 0$, where $\gamma = \sqrt{j\omega\mu(\sigma+j\omega\varepsilon)} = \alpha + j\beta$
α: attenuation constant,
β: phase constant, wavenumber,
γ: propagation constant
skin depth: $\delta = \frac{1}{\alpha}$
intrinsic wave impedance: $\eta = \sqrt{\frac{j\omega\mu}{\sigma+j\omega\varepsilon}}$
Uniform plane wave solution: $E_x(z) = Ae^{-\gamma z} + Be^{\gamma z}$

Phase velocity: $v_p = \frac{\omega}{\beta}$
Group velocity: $v_g = \frac{\mathrm{d}\omega}{\mathrm{d}\beta}$
Normal dispersion: $\frac{dv_p}{d\omega} < 0$
Abnormal dispersion: $\frac{dv_p}{d\omega} > 0$

Polaroid: Let one linear polarization (LP) pass and block

the other orthogonal LP.
Quarter-wave plate: The wave component polarized along the fast axis exceeds that along the slow axis by 90°.

Uniform plane wave reflection and transmission at a flat interface with normal incidence:
Reflection coefficient: $\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$
Standing wave ratio: SWR $= \frac{1+|\Gamma|}{1-|\Gamma|}$
Surface resistance of a conductor: $R_s = \sqrt{\frac{\pi f \mu}{\sigma}}$
Conductive loss per unit area: $P_a = \frac{1}{2}|\boldsymbol{J}_s|^2 R_s$
Radiation Pressure: $P_{\text{rad}} = 2\boldsymbol{S}_i \frac{1}{c}$
Momentum density: $\boldsymbol{p} = \frac{\boldsymbol{S}}{c^2} = \mu_0 \varepsilon_0 \boldsymbol{S}$
Momentum conservation in field-particle interaction:
$\nabla \cdot \overline{\overline{\sigma}} = \boldsymbol{f} + \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{S}}{\partial t}$ where
$\boldsymbol{f} = \rho \boldsymbol{E} + \boldsymbol{J} \times \boldsymbol{B}$,
$\overline{\overline{\sigma}} = \boldsymbol{E}\boldsymbol{D} + \boldsymbol{B}\boldsymbol{H} - \frac{\boldsymbol{E}\cdot\boldsymbol{D} + \boldsymbol{B}\cdot\boldsymbol{H}}{2}\overline{\overline{I}}$

Transmission Line Theory

Transmission Line Equation in frequency domain:

$$\frac{\mathrm{d}V}{\mathrm{d}z} = -(j\omega L + R)I, \quad \frac{\mathrm{d}I}{\mathrm{d}z} = -(j\omega C + G)V$$

General solution: $V(z) = Ae^{-\gamma z} + Be^{\gamma z}$
Propagation constant: $\gamma = \alpha + \mathrm{j}\beta = \sqrt{(R + \mathrm{j}\omega L)(G + \mathrm{j}\omega C)}$
Characteristic impedance: $Z_0 = \sqrt{\frac{R + \mathrm{j}\omega L}{G + \mathrm{j}\omega C}}$

For lossless TL:
General solution in coordinate ($d = \ell - z$) referenced at the load: $V(z) = V^+(e^{j\beta d} + \Gamma_L e^{-j\beta d}) = V^+ e^{j\beta d}(1 + \Gamma_L e^{-j2\beta d})$
Line impedance: $Z(z) = Z_0 \frac{Z_L + jZ_0 \tan(\beta\ell)}{Z_0 + jZ_L \tan(\beta\ell)} = Z_0 \frac{1+\Gamma(z)}{1-\Gamma(z)}$
Line reflection coefficient: $\Gamma(z) = \frac{Z(z) - Z_0}{Z(z) + Z_0} = \Gamma_L e^{-2j\beta\ell}$
For quarter wave transformer: $Z_{\text{in}} = \frac{Z_0^2}{Z_L}$, $I_L = -j\frac{V_{\text{in}}}{Z_0}$, $V_L = -jI_{\text{in}} Z_0$

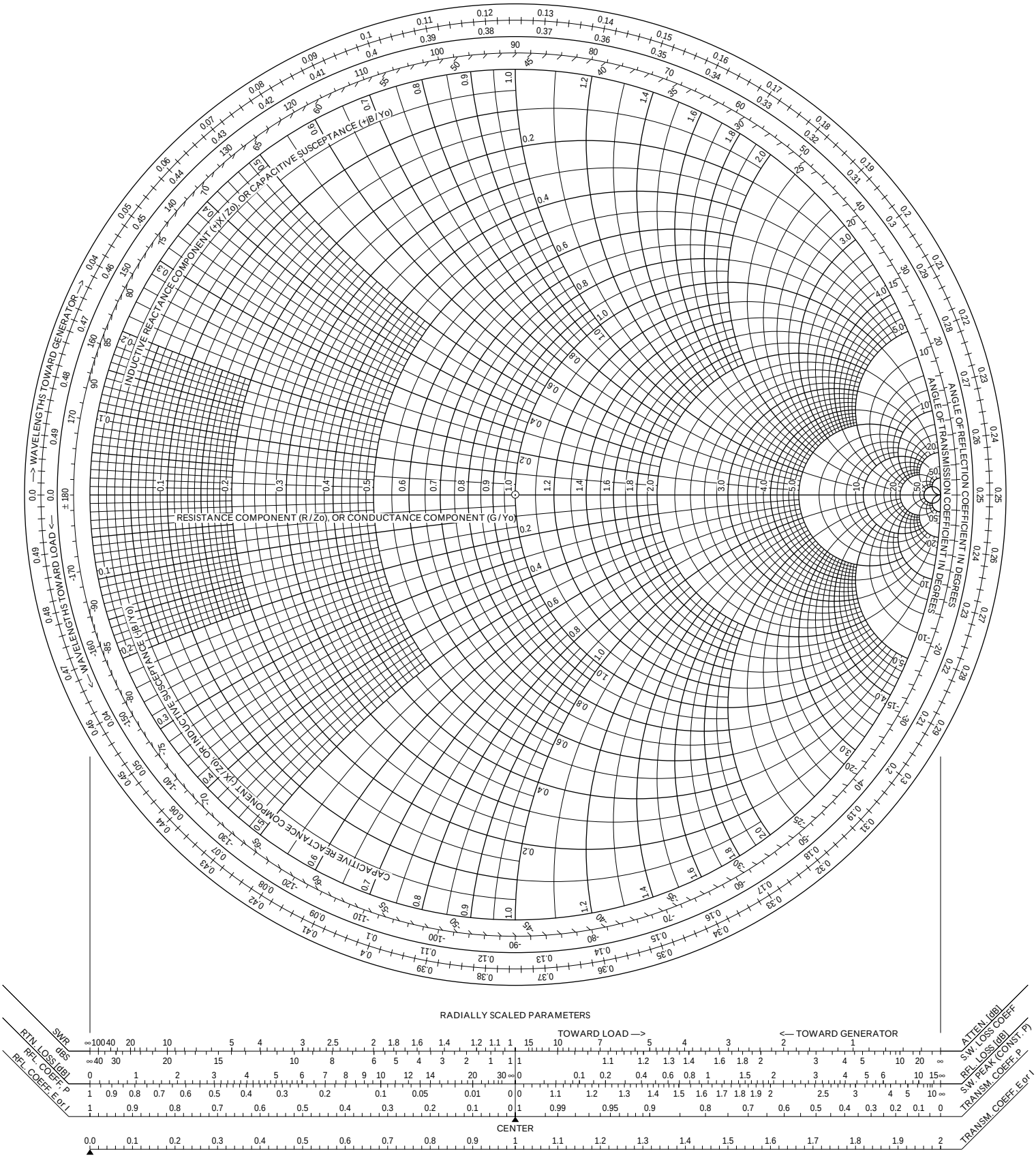
d'Alembert wave solution: $V(z,t) = Af(t - \frac{z}{v}) + Bf(t + \frac{z}{v})$
Round trip equation: $V^+(0,t) = \tau_s f(t) + \Gamma_S \Gamma_L V^+(0, t - 2T)$

Smith Chart:
Equal r circle: $(\Gamma_r - \frac{r}{r+1})^2 + \Gamma_i^2 = (\frac{1}{r+1})^2$
Equal x circle: $(\Gamma_r - 1)^2 + (\Gamma_i - \frac{1}{x})^2 = (\frac{1}{x})^2$

Useful Constants

free space permittivity: $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m
free space permeability: $\mu_0 = 4\pi \times 10^{-7}$ H/m
free space light speed: $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 3 \times 10^8$ m/s
free space intrinsic wave impedance: $\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} \approx 120\pi \approx 376.73\ \Omega$
elementary charge: $e = 1.602 \times 10^{-19}$ C
electron mass: $m_e = 9.109 \times 10^{-31}$ kg

Smith Chart



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